



Adroit Certified Engineer Exam

User Requirements Specification (URS)



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Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa
adroitscada.com

Version	Date	Author	Comment	Approval
0.0	24.02.2020	A. Remmers	Original	
1.0	10.06.2020	J. Duncan	Update Front Cover	
2.0	21.10.2020	D. Wibberley	Updated document	
3.0	22.10.2020	A. Remmers	Grammar fix	
4.0	26.10.2020	J. Duncan	Finalise	
5.0	26.10.2020	J. Duncan	Minor grammar corrections	
6.0	10.11.2020	J. Duncan	Update after feedback and comments	
7.0	23.11.2020	J. Duncan	Corrections and update	
8.0	22.05.2024	J. Duncan	Change to Juice Plant and minor corrections	
9.0	04.07.2024	D. Gibson	Review	
10.0	25.06.2025	J. Duncan	Finalise and QMS	



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1. Adroit Certified Engineer Exam

The purpose of this exam is to test your competency in configuring the Adroit SCADA products using various additional components. Therefore, we have designed the exam around a real plant and a typical user's requirements and functional design specifications.

In the real world, engineers have different approaches and philosophies when it comes to engineering a project. We recommend that you follow our Best Practices guide to engineer your solution.

Keep in mind that default installation folders are structured in a specific way for a reason and you should strive to adhere to them wherever possible to avoid loss of data during backup procedures.

The outcome of this exam is that, should you be successful, we will feel very comfortable in recommending you to any of our customers. We also believe that you will be able to charge a premium for your services in the field. If you are working for an end customer, then we feel that your organisation will be in very good hands. This is not an easy task, but we believe you will be justifiably proud once you successfully complete it.

All of Adroit's software products, drivers, knowledge base, help files and how to videos are available from the website <https://adroitscada.com/>

2. Prerequisites

It is advised that you attend the Adroit SmartUI Basic, Adroit SmartUI Advanced, and Report Suite and Alarm Management courses before attempting this exam, however there is no substitute for experience, and anyone can attempt the exam anytime.

We recommend that you familiarise and even base your exam project on our Best Practises, the sample Standards and FDS/DRP Documents that can be found on our website or obtained from our Support Department.

The Certified Engineers Exam Documentation:

- This document, the Certified Engineer Exam User Requirement Specification (URS).
- Adroit Certified Engineer Exam - Disaster Recovery Procedure (DRP) Template.
- Adroit Certified Engineer Exam - Function Design Specification (FDS) Template.
- Adroit Certified Engineer Exam - Supporting Excel Spreadsheets:
 - I/O Schedule,
 - Problem Solving Tasks, Marking Sheet and Summary.

3. The User Requirements Specification (URS)

3.1. Project Requirements

ABC Spices are in need of a simple batching system to produce their range of Juices for use in the food industry. The plant is simple since their juices consist of water mixed with two ingredients for a specified amount of time. These juice recipes need to be created and stored in a SQL Express database. The system needs to enable the operator to select a juice to be manufactured and to then download the target values for that recipe from the database to the Adroit Batch OPC Server on which the system is based (This sample Batch OPC Server is installed with any Adroit system).

The Database needs to contain the following fields:

- Product Name,
- Target Mass Ing A,
- Target Mass Ing B,
- Target Mass H₂O,
- Target Temperature and
- Mixing Time

The company makes four recipes/products namely; Apple Juice, Peach Juice, Pear Juice and Grape Juice. The batching plant consists of a single mixing bin which has a capacity of 1000 kg in which the two required ingredients are added automatically once the batch is started from within the simulation Batch OPC Server.

The Batch OPC Server has been designed to simulate PLC operation. Water is dispensed from a pump automatically, again, controlled by the Batch OPC Server.

It is important that the plant runs with a hot standby Agent Server configuration and has one remote client connected for remote viewing purposes.

DW Asked the following question.

What licensing do I need for an Adroit with 30 scan points and runs in a hot standby Agent Server configuration and has one remote client connected for remote viewing purposes?

It is critical that we keep a record of each batch by inserting the relevant information from each batch into the SQL database. You will need to ensure that each batch record contains the following information:

- Batch Number
- Batch Start Time & Batch Stop Time ensuring that the record ends up in SQL as a Date & Time
- Product Name
- Target Mass for Ing A (kg), Ing B (kg), H₂O (Water (l)), Mixing Time (sec)
- Actual Mass for Ing A (kg), Ing B (kg), H₂O (Water (l)), Mixing Time (sec)
- Comment on the Batch entered by the Operator

What Adroit Agent should I use to send my Batch data to a SQL Server database and do I need a license for this functionality?

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The operator's interface should be uncluttered, pleasing and navigation between screens must be clear and logical. Navigation should be achieved by the use of templates wherever possible to ensure that maintenance is kept to a minimum. Templates should also be used wherever it is necessary to use common screens, such as maintenance and running hour screens.

What is the best practise suggestion for designing an uncluttered UI in Adroit SCADA?

Is there a best practises document I can use to help me design the operator interface?

The plant needs to be run in both Automatic and Manual mode. Selection between these operation modes must be performed by using a switch on the Control/Management Graphic Form. This plant status should be displayed on the Operator/Plant Graphic Form in addition to the Control/Management Graphic Form, the change to Manual can only be performed by Engineers or Electricians, not by Operators.

The Automatic mode will require the plant to be in a "Stopped" state, which can be reset from a button on the control screen thus resetting the entire plant. The operator needs to select a product to be manufactured and then press a button to "Start" the process.

The status of the PLC will then change to "Batching". The details of the required recipe must then be loaded and clearly displayed in a logical format. The required information includes the "target" masses of the ingredients as well as the required mixing time and temperature. Other information is available and should be utilised and presented to the Operators logically. The currently logged on user must be displayed through the use of Graphic Form scripting. (There is a knowledge base article on how to achieve this in the Adroit Knowledge Base which can be found on the [website](#)).

Once the target masses are generated by the simulation Batch OPC Server, it will start the "Mixing" phase and change the status integer accordingly. This integer is found in the I/O schedule attached. This will start an Agitator/Mixer and you will need to display the start and end times for the batch, the current status of the plant as well as all the other states of the physical plant.

How do I setup the OPC Driver in Adroit?

Once the Mixing is complete, the "Heating" phase will begin where the mixture will be heated to a predefined temperature as set by the operator. Once the heating phase is complete, the valve under the mixing tank will open automatically and the mix will discharge at the set rate.

The status integer at this stage will be changed to reflect "Discharging". Once empty, the status will change to "Batch Complete" and a button to allow the Operator to commit the records to the database should become enabled or visible.

The complete record of the Batch must then be written to a database, including a comment field in which the Operator can enter some free text, prior to pressing a button to update the batch record database. Once the transaction is complete you will need to reset the plant, setting it into the "Stopped" state once again. On a Graphic Form, the Operator needs to be able to select a date/time range and return and display all records associated with that selection.

A Functional Design Specification document must be submitted. (as per our attached template)

I need an FDS document example for my project, where can I find it?

3.2. Navigation and Screen Layout

Using a TGO NavToolbar on the bottom quarter of the Operator Screen you need to be able to navigate between the Trend screen, the Event Viewer and the Alarm Viewer, each replacing the other. The top 75% of the Operator Screen must be reserved for the graphical representation of the plant. The default layout/workspace must be the Operator screen with the Trend screen displayed. The navigation should be done through a menu on the left and a TGO in which the Graphic Forms will be loaded.

How do I use the TGO NavToolbar control in a graphic form ?

3.3. Maintenance and MIS

In the event of a catastrophic SCADA system failure, a Disaster Recovery Procedure (DRP) document must exist detailing the steps to follow to recover the entire system from a backup, as well as hardware requirements and any other relevant information. This document needs to be submitted with the rest of the exam.

What sort of Adroit information would I need in a Disaster Recovery Procedure document for an Adroit SCADA system?

In order to give the maintenance staff additional assistance with each motor, the electrician must belong to his own Windows security group. It is also important that the running hours of each pump and the number of times that it has been stopped and started (in either Auto/Man) is recorded via a second 'running stats' template that must be opened by way of an Execute command behaviour triggered from a mouse hover event on the text describing of the motor on the Operator Screen.

What Agent would I use to track the running hours of a pump?

The mixing motor and the water pump should be able to be operated in Auto/Manual mode from a template screen that can only be accessed from the running stats Graphic Form. The running totals for each motor can only be reset by an "Engineer" with Administrator privileges. However, you also need to consider the following requirements.

The Control/Management Graphic Form must display the following information:

- The Adroit Cluster name.
- Which Agent Server is the Master
- The average number of putslots happening in the system.
- The HASP serial number.
- The status of the Device health (OPC Server).
- Adroit recommends that you always have at least 25% spare capacity on the memory usage on the SCADA PC. Therefore display the PC Memory Usage. This value must be alarmed accordingly.

What Agent would give me details such as the Adroit Cluster Name and the HASP Serial number?

The status of the device needs to be displayed on the control screen and should the OPC Server fail or change from "Healthy" state, as contained in the device Agent header slots, an alarm must be raised.

[What Agent and Slot would give me details of the health of an OPC Driver?](#)

3.4. Security

Using Windows local security create three levels of user groups and named users.

1. Operators/Operator

- a. Can operate the plant but cannot access the Running and Maintenance templates, can reset the plant.

2. Electricians/Electrician

- a. Can access everything the Operators can as well as access the Running and Maintenance templates, but they cannot reset the running statistics of the motors.

3. Engineers/ Engineer (Administrator)

- a. Can access everything the Operators/Electricians can as well as access the Running and Maintenance templates, and they can reset the running statistics of the motors.

As it is not possible to submit your user and group structure with the Adroit configuration backup, you are required to add the Windows security procedures to your Disaster Recovery Procedure document. Please provide only the details of the user and group configuration from a Windows point of view. The Adroit specific security settings are included with the project backup and will be marked accordingly.

[How do I setup security in the Adroit SCADA Server and Operator?](#)

3.5. Logging Strategy

All Operational data should be logged using Adroit's proprietary datalogging subsystem for a maximum period of 30-days at intervals of 1 sec.

DBLogs to be configured to store long term raw process data. Use DBQuery to log the following to a user defined table.

- Setpoints
- Process Variables
- Product Name
- Batching data (start time, end time, qty etc)

All tags that require operator actions need to be audited and sent to the SQL Express database.

[How do I use the DBQuery Agent in Adroit?](#)

3.6. Alarming Strategy

The capacity of the mixing tank is 1000kg and must generate an alarm when the total mass in the tank exceeds 80% and then 90% of the capacity, which is the sum of all the batched ingredients and water. As with certain real-world situations the maximum level of the mixing vessel can vary depending on the current batch. Thus statically assigning the alarm limits is not practical, so these should vary based on the engineering maximum value. The calculation of the Hi and Hihi alarm limits needs to be done in a VBScript. The tank empty alarm must be low priority and doesn't require acknowledgement by the Operator, therefore, is classified as an Event.

It is required that the navigation button/object on the AlarmViewer page on the TGONavToolbar displays the number of unacknowledged alarms in the system and flash red if there are any unacknowledged alarms.

[What Agent and Slot in Adroit will give me the number of unacknowledged alarms?](#)

All alarms need to be routed to an Historical Alarm Database using the AlarmManagement Agent and displayed on a dedicated modeless graphic form using the standard Historical Alarm Viewer control in the Operator, but also available in the Report Suite.

[How do I log alarms into a SQL Database?](#)

Every time the plant is reset, you are required to get an operator confirmation and write this event to the Audit database available in Adroit.

[How do I get a confirmation from a button action and log this to an auditing database in adroit?](#)

[How do I log an event in the Operator to a database for auditing purposes?](#)

The alarms must be routed to an alarm list that can only be locally acknowledged by Operators.

[How do I route alarms to an Adroit Alarm list such that only local Operators can acknowledge these alarms?](#)

3.7. Support

Setup the Adroit Protocol Driver Monitor utility as well as the Scan Configuration Monitoring utility to monitor the Batch OPC Server you are running. Take note of the number of scan jobs you are running and how many tags are in each scan job. Submit a screen grab of the results as part of your exam submission.

[How do I set up the Adroit Protocol Monitor?](#)

Use the Adroit Tag Monitoring utility to monitor the status of the Driver Device Alarm, Running and Healthy Slots. Save this configuration in case you would like to monitor this at some other stage. Submit a screen grab of the configuration as part of your exam submission.

[How do I use the Adroit Tag Monitor application?](#)

Use the Adroit Health Monitor Service and attach the latest Performance log for the Agent Server as you would do when submitting a support request.

3.8. Backups and Version Control

Using the Adroit Backup function, schedule a weekly custom backup that includes your Adroit Project configuration and the SQL database every Monday at 00h00.

The Agent Server should be automatically saved every 30 seconds (in reality you need to decide whether it is required and what "granularity" is required) and also ensure that that Counter Agents are set to be "memory retentive" (in case of a power failure that could shut the AS down suddenly). Display both the set time and time left, as well as the SaveNow slot status on the Control/Management Graphic Form using the Timer Agent.

[How do I set Adroit up so that the WGP file saves itself every 30 seconds?](#)

Configure Adroit to enable the configuration engineers to add version description entries in the WGP file every time it is saved/edited.

3.9. Architecture and General IT

There are many standards when it comes to a tag naming convention. You are required to use the S95 Hierarchy to create a tag naming convention, typically:

Country, Region, Enterprise, Site Name, Area, Area Type, Work Centre, Work Centre Type, Equipment, Equipment Type, Control Module, Control Module Type, Signal, Signal Type, Signal Process, Signal Measurement.

Set up an Active Cluster configuration. In order to submit your Cluster configuration to us, add a section to the Disaster Recovery Procedure document. Both the Agent Server as well as the UI configuration must be taken into account. You are required to give descriptions of all relevant settings within the Adroit setup as well as all configuration applying to the Servers and UIs (both from an Adroit as well as a Windows perspective).

The PLC you will be using for this Exam is the simulation OPC Server, called Batch OPC Server and will simulate the operation of the Plant.

Using the Adroit Excel Add-In tool, create and scan all the Agents as per provided IO schedule, including alarming configuration. Save a copy and submit together with the rest of the files.

3.10. Additional Requirements

Once all the motors have been added and their Running Stats template have been successfully created, use the Agitator motor and convert it into an Object Model Template using the Object Model so that additional motors can be added when needed. Then use this template to add an additional 2 motors to the plant.

[How do I convert a graphic of a motor into an object within adroit?](#)

Install Performance Anywhere and display the temperature trend and status of the plant.

Additional marks are available should you also implement the Adroit Air Mobile datasource and App. In which the alarms are sent, and at the end of any Batch the process variables are also sent to the App.

4. I/O Schedule

Refer to the supplied I/O Schedule [ATQMS-MKTG-EXL-Adroit Certified Engineer Exam_IO Schedule_2025]

5. Problem Solving Tasks

Refer to the supplied spreadsheet ATQMS_MKTG_EXL_Adroit Certified Engineer Exam_2025 and complete the section (Tab) "Problem Solving".

6. Submission

The following can be handed in on a disc or other storage device at our premises:

1. A full Adroit Backup (using the Adroit backup functionality in Adroit Config Editor) including registry entries. Check that your backup includes all the files you have worked on.
Note: Should you elect to submit your project on VMware, we will not be able to test the restore functionality required.
2. The written question part of the exam "Problem Solving Tasks" as per attached spreadsheet.
3. A Functional Design Specification (FDS) document.
4. Disaster Recovery Procedure (DRP) document which the marker will use to set up the system for marking.
5. SQL Express database as well as a section in the Disaster Recovery Procedure document on how to restore it.
6. Any other relevant files requested screen grabs and/or procedures.